

## Problem Statements

*Choose a challenge. Build a solution. Make an impact.*

### **MEDIUM** AI Code Review & Security Assistant

**Problem Statement:** Developers can accidentally introduce bugs and security vulnerabilities while writing code. Finding these problems early can prevent larger issues later. Build an AI-powered code review tool that analyzes code, identifies potential problems, and suggests improvements.

**Your solution should:**

- Allow users to enter or upload code.
- Identify possible bugs or security vulnerabilities.
- Explain detected issues in simple language.
- Suggest a safer or improved version of the code.
- Assign a severity level where possible.

**Expected Outcome:** An AI assistant that helps developers write safer and better-quality code.



### GENERAL GUIDELINES FOR TEAMS

- Teams may choose any programming language, framework, database, ML model, or development stack suitable for their solution.
- The problem statement describes the goal; teams are free to decide how to implement the solution.
- A working prototype is preferred over a concept-only submission.
- Teams should clearly demonstrate the problem, their approach, and the impact of their solution.
- For data-driven challenges, teams may use suitable public datasets and clearly mention their sources.
- For AI-based solutions, teams should be able to explain the role of AI in their solution.